Time evolution operator

We seek a time-evolution operator $U(t,t_0)$ such that

$$\psi(x,t) = U(t,t_0)\psi(x,t_0)$$

Start with the Schrodinger equation.

$$i\hbar\frac{\partial}{\partial t}\psi(x,t) = H(t)\psi(x,t)$$

Substitute for $\psi(x,t)$.

$$i\hbar\frac{\partial}{\partial t}\Big[U(t,t_0)\psi(x,t_0)\Big] = H(t)U(t,t_0)\psi(x,t_0)$$

Expand the left-hand side by the product rule.

$$i\hbar\frac{\partial}{\partial t}U(t,t_0)\psi(x,t_0) + i\hbar U(t,t_0)\frac{\partial}{\partial t}\psi(x,t_0) = H(t)U(t,t_0)\psi(x,t_0)$$

Noting that

$$\frac{\partial}{\partial t}\psi(x,t_0) = 0$$

we have

$$i\hbar\frac{\partial}{\partial t}U(t,t_0)\psi(x,t_0) = H(t)U(t,t_0)\psi(x,t_0)$$

By operator equivalence

$$i\hbar\frac{\partial}{\partial t}U(t,t_0) = H(t)U(t,t_0)$$

Rewrite as

$$\frac{\partial}{\partial t}U(t,t_0) = -\frac{i}{\hbar}H(t)U(t,t_0)$$

Rearrange as

$$\frac{1}{U(t,t_0)}\,\partial U(t,t_0) = -\frac{i}{\hbar}H(t)\,\partial t$$

Integrate both sides.

$$\int_{t_0}^{t}\frac{1}{U(\tau,t_0)}\,dU(\tau,t_0) = -\frac{i}{\hbar}\int_{t_0}^{t}H(\tau)\,d\tau$$

Solve the first integral.

$$\log U(t,t_0) - \log U(t_0,t_0) = -\frac{i}{\hbar}\int_{t_0}^{t}H(\tau)\,d\tau$$

By $U(t_0,t_0) = 1$ the second log vanishes.

$$\log U(t,t_0) = -\frac{i}{\hbar}\int_{t_0}^{t}H(\tau)\,d\tau$$

Hence

$$U(t,t_0) = \exp\left(-\frac{i}{\hbar}\int_{t_0}^{t}H(\tau)\,d\tau\right) \tag{1}$$

Eigenmath script